

Nonparametric Statistics

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통계학과

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Motivation

- ① What are Parametric Statistics and Nonparametric Statistics?
- ② Why do we need to know **Nonparametric Statistics**?
- ③ What kind of Nonparametric Statistics exists?

1. Parametric Statistics

Parametric statistics assumes that the population follows a specific probability distribution characterized by a finite number of parameters. Statistical inference is then performed by estimating or testing these parameters. For example, if

$$X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2),$$

the population distribution is completely determined by the two parameters μ and σ^2 .

1. Estimating the Mean of a Normal Distribution

Suppose students' exam scores are assumed to follow $X_i \sim N(\mu, \sigma^2)$. The sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

can be used to estimate the population mean μ .

2. Two-Sample t -Test

Suppose we want to compare the means of two populations. Under suitable distributional assumptions, we can test $H_0 : \mu_1 = \mu_2$ using a two-sample t -test.

2. Nonparametric Statistics

Nonparametric statistics does not assume that the population belongs to a specific finite-dimensional family of probability distributions. Instead, it allows the underlying distribution to have a more flexible and unspecified form.

The term nonparametric does not mean that there are no parameters at all. Rather, it means that the population distribution is not restricted to a model determined by only a fixed finite number of parameters.

Why Do We Need to Know Nonparametric Statistics?

Nonparametric statistics is useful because real-world data do not always satisfy the assumptions required by parametric methods.

- **Fewer distributional assumptions:** We do not need to assume that the data follow a specific distribution such as the normal distribution.
- **Robustness:** Nonparametric methods are often less sensitive to outliers, skewed data, and model misspecification.
- **Small or unusual data sets:** They can be useful when the sample size is small or when the underlying distribution is difficult to identify.
- **Flexible modeling:** Nonparametric methods can capture complex structures that may not be well described by a fixed parametric model.

In short, nonparametric statistics provides useful alternatives when parametric assumptions are unrealistic or too restrictive.

Introduction to Nonparametric Statistics

Introduction

Let $S_X = X_1, \dots, X_n \sim P_X$ and $S_Y = Y_1, \dots, Y_m \sim P_Y$ where

- X_i 's denote the response of individuals in the control group (from the population P_X)
- Y_i 's denote the response of individuals in the treatment group (from the population P_Y)
- n, m are the sample sizes

If there is no difference between the two populations (i.e., $P_X = P_Y$), the two-sample test problem is to test

$$H_0 : P_X = P_Y,$$

which is non-trivial. A simple alternative might be the t -test (or Z -test) with

$$H_0 : \mu_X = \mu_Y,$$

where μ_X, μ_Y are the mean of P_X and P_Y , respectively.

Introduction (Cont'd)

Testing the mean has a potential problem: presence of outliers $\rightarrow$ low power.

Consider a simple example:

$$S_X = -2, -1, -1, 0, -2, -1, 0, -1, -2, 100$$

$$S_Y = 6, 12, 10, 4, 13, 8, 7, 9, 11, 10$$

The sample averages of both samples are 9. Thus, the t -test or other mean test fail!

We will see a few new methods that are much more robust to outliers. During this seminar, we will consider two common scenarios in two sample tests: *paired sample* and *non-paired sample*.

Paired Sample: Sign Test

A common practice is to introduce a new set of variables $Z_1, \dots, Z_n$, where $Z_i = Y_i - X_i$ for each $i = 1, \dots, n$.

Under $H_0 : P_X = P_Y$, Z_i is from a distribution with mean 0 (and median 0).

The distribution of Z_i has median 0 $\rightarrow$ the chance of $Z_i < 0$ and the chance of $Z_i > 0$ are the same.

Thus, the number of Z_i 's being positive ($= T$) should follow a binomial distribution $\text{Bin}(n, 0.5)$ under H_0 .

A simple way to compute a p -value is to calculate prob. of observing a more extreme event against H_0 than what is observed.

Toy example

$$S_Z = -8, -13, -11, -4, -15, -9, -7, -10, -13, 90,$$

where $T = 1$ in our case. So a more extreme event is $T = 0$ and

$$P(T = 1) = \binom{10}{1} \frac{1}{2^{10}}, \quad P(T = 0) = \binom{10}{0} \frac{1}{2^{10}}.$$

However, $T \approx 10$ is also against H_0 . Thus, the p -value will be

$$p \text{ value}_1 = 2(P(T = 1) + P(T = 0)) = \frac{22}{1024} \approx 2.15\%.$$

Paired Sample: Sign Test

Asymptotic normality:

$\text{Bin}(n, 0.5)$ is approximately normal with $N\left(\frac{n}{2}, \frac{n}{4}\right)$.

Thus, we can use

$$T^* = \frac{T - n/2}{\sqrt{n/4}},$$

with the following p -value

$$p \text{ value}_2 = 2P(N(0, 1) \geq |T^*|) \approx 1.14\%.$$

The former one is the exact p -value whereas the later one is an asymptotic approximation.

Convergence in Probability

Definition

Let $\{X_n\}$ be a sequence of random variables and let X be a random variable defined on a sample space Ω . We say that X_n **converges to probability** to X if for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0.$$

We denote it as $X_n \xrightarrow{p} X$.

Example

Let $X_1, \dots, X_n \sim \text{Ber}(p)$, $0 < p < 1$. Then, $\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} p$.

Weak Convergence

Definition

Let $\{X_n\}$ be a sequence of random variables and let X be a random variable. Let F_n and F be the cdfs of X_n and X , respectively. We say that X_n **converges to distribution** to X if

$$\lim_{n \rightarrow \infty} F_n(x) = F(x), \quad \forall x \in C(F)$$

where $C(F)$ is the set of all points where F is continuous.

We denote it as $X_n \xrightarrow{d} X$.

Example

Let $T_n \sim t_n, n = 1, 2, \dots$. Then, $T_n \xrightarrow{d} Z$ where $Z \sim N(0, 1)$.

We call **converges to distribution** as **weak convergence**.

Paired Sample: Signed-Rank Test

A little bit more assumption $\rightarrow$ the distribution of Z_i is symmetric.

Let $\text{sgn}(Z_i)$ denote the sign of Z_i (+ or -). And define

$$R_i = \text{rank}(|Z_i|),$$

where $R_i = 1$ denotes the smallest $|Z_i|$'s.

The signed-rank test uses a test statistic (signed-rank sum)

$$W = \sum_{i=1}^n \text{sgn}(Z_i) R_i.$$

Under H_0 , the median of Z_i is 0, so W should be close to 0 (test if the median is 0 or not).

And the variance of W is

$$\text{Var}(W) = \frac{n(n+1)(2n+1)}{6}, \text{ and } \frac{W}{\sqrt{\text{Var}(W)}} \xrightarrow{d} N(0, 1).$$

Paired Sample: Signed-Rank Test

The p -value can be computed using

$$\begin{aligned} p \text{ value} &= 2P \left(N(0, 1) \geq \left| \frac{W - 0}{\sqrt{\text{Var}(W)}} \right| \right) \\ &= 2P \left(N(0, 1) \geq \sqrt{\frac{6W^2}{n(n+1)(2n+1)}} \right). \end{aligned}$$

Toy example

Z_i	-8	-13	-11	-4	-15	-9	-7	-10	-13	90
$\text{sgn}(Z_i)$	-	-	-	-	-	-	-	-	-	+
R_i	3	7.5	6	1	9	4	2	5	7.5	10

$W = -35$, leading to a p -value

$$2P \left(N(0, 1) \geq \sqrt{\frac{6W^2}{n(n+1)(2n+1)}} \right) \approx 7.45\%.$$

Thus,

Non-paired Sample: Rank-Sum Test

When two samples are not paired, we can no longer use the sign test or the signed-rank test. The rank-sum test directly test the null hypothesis $H_0 : P_X = P_Y$.

Idea: pull both samples together and compute the rank of each observation.

- Again, two samples are $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$.
- Combine them together to form a new data $D_1, \dots, D_{n+m}$, where $D_i = X_i$ for $i = 1, \dots, n$ and $D_i = Y_{i-n}$ for $i = n+1, \dots, n+m$.
- Let $S_1, \dots, S_{n+m}$ be the rank of $D_1, \dots, D_{n+m}$.

First consider the following preliminary statistics:

$$Q_1 = \sum_{i=1}^n S_i, \quad Q_2 = \sum_{i=n+1}^{n+m} S_i.$$

Note that

$$Q_1 + Q_2 = \frac{(n+m)(n+m+1)}{2}.$$

Non-paired Sample: Rank-Sum Test

Define two statistics:

$$U_1 = Q_1 - \frac{n(n+1)}{2}, \quad U_2 = Q_2 - \frac{m(m+1)}{2}.$$

We can use any of them as our test statistic. Each U_I has several nice properties:

$$\mathbb{E}(U_I) = \frac{nm}{2}, \quad \text{Var}(U_I) = \frac{nm(n+m+1)}{12},$$

and

$$\frac{U_I - \mathbb{E}(U_I)}{\sqrt{\text{Var}(U_I)}} \xrightarrow{d} N(0, 1).$$

The p -value will be

$$\begin{aligned} p \text{ value} &= P \left(|N(0, 1)| \geq \left| \frac{U_I - \mathbb{E}(U_I)}{\sqrt{\text{Var}(U_I)}} \right| \right) \\ &= P \left(|N(0, 1)| \geq \left| \frac{U_I - nm/2}{\sqrt{nm(n+m+1)/12}} \right| \right). \end{aligned}$$

Summary for Sign, Signed-Rank, and Rank-Sum Test

Assumptions (Sign Test)

- The data is paired.
- Every pair of observations is IID from an unknown continuous (or ordinal) distribution function.

Assumptions (Signed-Rank Test)

- The data is paired.
- Every pair of observations is IID from an unknown continuous distribution function (the distribution of Z_i is symmetric).

Assumptions (Rank-Sum Test)

- All observations are independent from each other.
- Observations within the same sample are IID.

The rank-sum test requires much weaker assumptions compared to the signed-rank or signed tests. And it works for ordinal data.

CDF and EDF

CDF: Cumulative Distribution Function

For a r.v. X , its CDF $F(x)$ contains all the probability structures of X . Here are some properties of $F(x)$:

- (probability) $0 \leq F(x) \leq 1$.
- (monotonicity) $F(x) \leq F(y)$ for every $x \leq y$.
- (right-continuity) $\lim_{x \rightarrow y^+} F(x) = F(y)$, where $y^+ = \lim_{\epsilon > 0, \epsilon \rightarrow 0} y + \epsilon$.
- $\lim_{x \rightarrow -\infty} F(x) = F(-\infty) = 0$.
- $\lim_{x \rightarrow \infty} F(x) = F(\infty) = 1$.
- $P(X = x) = F(x) - F(x^-)$, where $x^- = \lim_{\epsilon < 0, \epsilon \rightarrow 0} x + \epsilon$.

CDF: Cumulative Distribution Function

Why do we care about the CDF?

- CDF is the actual quantity that defines the probability structure of a random variable.
- The PDF exists only when the RV is continuous.
- The PMF exists when the RV is discrete.
- But CDF always exists – it is a unified quantity regardless of the RV being continuous or discrete.
- Moreover, there are cases where neither PDF nor PMF exist.

Example: X is a random variable such that with a probability of 0.5, it is from a uniform distribution over the interval $[0, 1]$ and with a probability of 0.5 it takes a fixed value 0.5.

EDF: Empirical Distribution Function

Given a value x_0 , $F(x_0) = P(X_i \leq x_0)$ for every $i = 1, \dots, n$. Namely, $F(x_0)$ is the probability of the event $\{X_i \leq x_0\}$.

A natural estimator of a probability of an event is the *ratio of such an event in our sample*.

$$\hat{F}_n(x_0) = \frac{\text{number of } X_i \leq x_0}{\text{total number of observations}} = \frac{\sum_{i=1}^n I(X_i \leq x_0)}{n}.$$

For every x_0 , we can use such a quantity as an estimator, so the estimator of the CDF, $F(x)$, is $\hat{F}_n(x)$.

This estimator, $\hat{F}_n(x)$, is called the *empirical distribution function (EDF)*.

Property of EDF

Let $Y_i = I(X_i \leq x)$. Then,

$$Y_i = \begin{cases} 1, & \text{if } X_i \leq x, \\ 0, & \text{if } X_i > x. \end{cases}$$

So, Y_i is a Bernoulli r.v. with a parameter

$$p = P(Y_i = 1) = P(X_i \leq x) = F(x).$$

That is, $Y_i \sim \text{Ber}(F(x))$. This implies

$$\mathbb{E}(I(X_i \leq x)) = \mathbb{E}(Y_i) = F(x)$$

and

$$\text{Var}(I(X_i \leq x)) = \text{Var}(Y_i) = F(x)(1 - F(x)),$$

for a given x .

Property of EDF

Recall that

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x).$$

- At each x , $\hat{F}_n(x)$ is an *unbiased estimator* of $F(x)$:

$$\text{bias}(\hat{F}_n(x)) = E(\hat{F}_n(x)) - F(x) = 0.$$

- The variance converges to 0 when $n \rightarrow \infty$. By definition of convergence in probability,

$$\hat{F}_n(x) \xrightarrow{p} F(x).$$

i.e., $\hat{F}_n(x)$ is a *consistent estimator* of $F(x)$.

- For a given x , by **Central Limit Theorem**,

$$\sqrt{n}(\hat{F}_n(x) - F(x)) \xrightarrow{d} N(0, F(x)(1 - F(x))).$$

i.e., $\hat{F}_n(x)$ is asymptotically normally distributed.

Remark

The previous result shows that for a given x ,

$$\left| \hat{F}_n(x) - F(x) \right| \xrightarrow{P} 0.$$

This is related to the pointwise convergence in mathematical analysis. We can extend this to a uniform sense (uniform convergence):

$$\sup_x \left| \hat{F}_n(x) - F(x) \right| \xrightarrow{P} 0.$$

For any $\epsilon > 0$,

$$P \left(\sup_x \left| F(x) - \hat{F}_n(x) \right| > \epsilon \right) \leq 2e^{-2n\epsilon^2}.$$

(Dvoretzky–Kiefer–Wolfowitz (DKW) Inequality)

Goodness-of-fit Test: One-sample Test

Before we describe the two-sample test, we first consider a simpler case, one-sample test, testing against a known distribution.

Test if $X_1, \dots, X_n$ are from a known distribution F_0 (goodness-of-fit test), i.e., $H_0 : X_1, \dots, X_n \sim F_0$.

- Kolmogorov-Smirnov test (KS test):

$$T_{KS} = \sqrt{n} \sup_x \left| \hat{F}_n(x) - F_0(x) \right|.$$

- Cramér-von Mises test:

$$T_{CM} = n \int \left(\hat{F}_n(x) - F_0(x) \right)^2 dF_0(x).$$

- Anderson-Darling test:

$$T_{AD} = n \int \frac{\left(\hat{F}_n(x) - F_0(x) \right)^2}{F_0(x)(1 - F_0(x))} dF_0(x).$$

Goodness-of-fit Test: Two-sample Test

Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ be the two samples with corresponding EDFs $\hat{F}_X$ and $\hat{F}_Y$. Here, we define the EDF by treating both sample as the same sample:

$$\hat{H}(t) = \frac{n\hat{F}_X(t) + m\hat{F}_Y(t)}{n+m} = \frac{1}{n+m} \left(\sum_{i=1}^n I(X_i \leq t) + \sum_{j=1}^m I(Y_j \leq t) \right).$$

- KS test:

$$T_{KS} = \sqrt{\frac{nm}{n+m}} \sup_x \left| \hat{F}_X(t) - \hat{F}_Y(t) \right|.$$

- Cramér-von Mises test:

$$T_{CM} = \frac{nm}{n+m} \int \left(\hat{F}_X(t) - \hat{F}_Y(t) \right)^2 d\hat{H}(t).$$

- Anderson-Darling test: $T_{AD} = \frac{nm}{n+m} \int \frac{\left(\hat{F}_X(t) - \hat{F}_Y(t) \right)^2}{\hat{H}(t)(1-\hat{H}(t))} d\hat{H}(t).$

Toy Example: Delivery Time

Suppose we observe the delivery times (in minutes) of the last 10 orders:

18, 22, 24, 25, 27, 29, 31, 35, 41, 48.

Let X denote the delivery time. The true CDF is $F(x) = P(X \leq x)$, but F is unknown. We estimate it using the empirical distribution function:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x).$$

For example,

$$\hat{F}_n(30) = \frac{\#\{X_i \leq 30\}}{10} = \frac{6}{10} = 0.6.$$

Thus, based on the observed data, we estimate that about 60% of deliveries are completed within 30 minutes.

Nonparametric Statistical Models and Inference

Histogram and Kernel Density Estimator

Histogram

Let $X_i \in [0, 1]$ and assume $p(x)$ is smooth and $|p'(x)| \leq L$ for all x . The histogram is to partition the set $[0, 1]$ into M bins and using the count of the bins as a density estimate:

$$B_1 = \left[0, \frac{1}{M}\right), B_2 = \left[\frac{1}{M}, \frac{2}{M}\right), \dots, B_M = \left[\frac{M-1}{M}, 1\right).$$

For a given point $x \in B_l$, the density estimator from the histogram is

$$\hat{p}_n(x) = \frac{\# \text{ of obs. within } B_l}{n} \times \frac{1}{\text{length of the bin}} = \frac{M}{n} \sum_{i=1}^n I(X_i \in B_l).$$

→ the ratio of observations within this bin is $\frac{1}{n} \sum_{i=1}^n I(X_i \in B_l)$, which should be equal to the density estimate times the length of the bin.

Histogram

For a given point $x \in B_l$, $\mathbb{E}(\hat{p}_n(x)) = \frac{F\left(\frac{l}{M}\right) - F\left(\frac{l-1}{M}\right)}{\frac{l}{M} - \frac{l-1}{M}} = p(x^*)$,

where $x^* \in \left[\frac{l-1}{M}, \frac{l}{M}\right]$. Then, the bias is

$$\text{Bias}(\hat{p}_n(x)) = \mathbb{E}(\hat{p}_n(x)) - p(x) \leq \frac{L}{M}.$$

→ the more bins we are using, the less bias the histogram has. Now, the variance is

$$\text{Var}(\hat{p}_n(x)) = M^2 \cdot \text{Var}\left(\frac{1}{n} \sum_{i=1}^n I(X_i \in B_l)\right) = M \cdot \frac{p(x^*)}{n} - \frac{p^2(x^*)}{n}.$$

→ the more bins we are using, the higher variance we are suffering.

Histogram

The MSE is

$$\text{MSE}(\hat{p}_n(x)) = \text{bias}^2(\hat{p}_n(x)) + \text{Var}(\hat{p}_n(x)) \leq \frac{L^2}{M^2} + M \cdot \frac{p_{\max}}{n} - \frac{p_{\max}^2}{n}.$$

we can choose “M”, the number of bins:

- Large M (undersmoothing): the first quantity (bias) will be small while the second quantity (variance) will be large.
- Small M (oversmoothing): the first quantity (bias) will be large, but the second quantity (variance) is small.

To balance the bias and variance, choose M that minimizes the MSE:

$$M_{\text{opt}} = \left(\frac{nL^2}{p_{\max}} \right)^{1/3}$$

Kernel Density Estimator

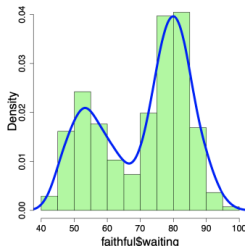


Figure: The KDE and the histogram of the faithful dataset in R.

The Kernel Density Estimator(KDE) is a function

$$\hat{p}_n(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right)$$

where $K(x)$ is called the kernel function that is generally a smooth, symmetric function and $h > 0$ is called the smoothing bandwidth.

Kernel Density Estimator

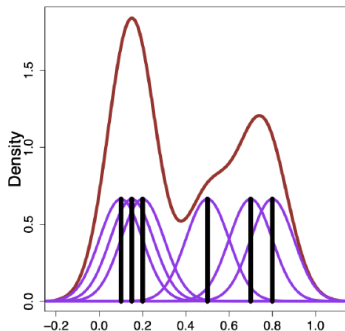


Figure: There are 6 data points located at where the black vertical segments indicate: 0.1, 0.2, 0.5, 0.7, 0.8, 0.15.

The KDE first smooth each data point into a purple density bump and then sum them up to obtain the final density estimate—the brown density curve.

Bandwidth and Kernel Functions

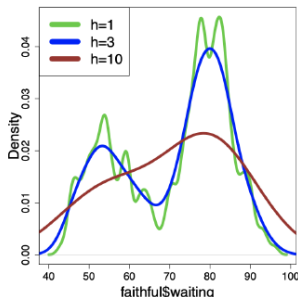


Figure: An example of applying different smoothing bandwidth h to the faithful dataset

- h is too small (undersmoothing): many wiggly structures
- h is too large (oversmoothing): the two bumps are smoothed out

Bandwidth and Kernel Functions

How about the choice of kernel function?

- $K(x)$ is symmetric.
- $\int K(x) dx = 1$: this requirement is needed to guarantee that $\hat{p}_n(x)$ is prob. density.
- $\lim_{x \rightarrow -\infty} K(x) = \lim_{x \rightarrow \infty} K(x) = 0$.

We usually use three kernel functions:

$$\text{Gaussian } K(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

$$\text{Uniform } K(x) = \frac{1}{2} I(-1 \leq x \leq 1),$$

$$\text{Epanechnikov } K(x) = \frac{3}{4} \max\{1 - x^2, 0\}.$$

Toy Example: Commuting Time

Suppose we collect the commuting times (in minutes) of 100 employees. We want to estimate the underlying density of commuting time. A parametric approach may assume $X \sim N(\mu, \sigma^2)$, but the true distribution may not be Gaussian. For example, there may be

- one group living close to the workplace,
- another group commuting from distant areas.

Thus, the distribution may have more than one peak. Using KDE,

$$\hat{p}_n(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right),$$

we can estimate the shape of the commuting-time distribution without assuming a specific parametric form.

Theory of the KDE: Bias

Assume that $X_1, \dots, X_n$ are IID sample from an unknown density function p .

For a given point x_0 , the bias of KDE is

$$\begin{aligned}\mathbb{E}(\hat{p}_n(x_0)) - p(x_0) &= E \left[\frac{1}{nh} \sum_{i=1}^n K \left(\frac{X_i - x_0}{h} \right) \right] - p(x_0) \\ &= \frac{1}{2} h^2 p''(x_0) \mu_K + o(h^2),\end{aligned}$$

where $\mu_K = \int y^2 K(y) dy$.

- $h \rightarrow 0$, the bias is shrinking at a rate $O(h^2)$.
- Bias will be very large at a point where the density function curves a lot.

Theory of the KDE: Variance

For the analysis of variance, an upper bound can be obtained by

$$\begin{aligned}\text{Var}(\hat{p}_n(x_0)) &= \text{Var}\left(\frac{1}{nh} \sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)\right) \leq \frac{1}{nh^2} E\left[K^2\left(\frac{X_i - x_0}{h}\right)\right] \\ &= \frac{1}{nh} p(x_0) \sigma_K^2 + o\left(\frac{1}{nh}\right),\end{aligned}$$

where $\sigma_K^2 = \int K^2(y) dy$. Therefore, the variance shrinks at rate $O\left(\frac{1}{nh}\right)$ when $n \rightarrow \infty$ and $h \rightarrow 0$.

At point where the density value is large, the variance is also large!

The MSE of the KDE:

$$\begin{aligned}\text{MSE}(\hat{p}_n(x_0)) &= \text{bias}^2(\hat{p}_n(x_0)) + \text{Var}(\hat{p}_n(x_0)) \\ &= \frac{1}{4}h^4|p''(x_0)|^2\mu_K^2 + \frac{1}{nh}p(x_0)\sigma_K^2 + o(h^4) + o\left(\frac{1}{nh}\right)\end{aligned}$$

$$\boxed{\text{MSE}(\hat{p}_n(x_0)) = O(h^4) + O\left(\frac{1}{nh}\right)}$$

Confidence Interval

Recall that

$$\hat{p}_n(x_0) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right) = \frac{1}{n} \sum_{i=1}^n Y_i,$$

where $Y_i = \frac{1}{h} K\left(\frac{X_i - x_0}{h}\right)$. Thus, the KDE evaluated at x_0 is actually a sample mean of $Y_1, \dots, Y_n$. By CLT,

$$\sqrt{n} \left(\frac{\hat{p}_n(x_0) - \mathbb{E}(\hat{p}_n(x_0))}{\sqrt{\text{Var}(Y_i)}} \right) \xrightarrow{D} N(0, 1).$$

When $h \rightarrow 0$, the asymptotic distribution of $\hat{p}_n(x_0)$ is

$$\sqrt{nh} (\hat{p}_n(x_0) - \mathbb{E}(\hat{p}_n(x_0))) \xrightarrow{D} N(0, p(x_0)\sigma_K^2).$$

Thus, a $100(1 - \alpha)\%$ CI can be constructed using

$$\hat{p}_n(x_0) \pm z_{1-\alpha/2} \sqrt{\frac{p(x_0)\sigma_K^2}{nh}}.$$

k -nearest neighborhood

k-Nearest Neighbor (kNN)

Let $X_1, \dots, X_n \sim p$ be a d -dimensional random sample (namely, $X_i \in \mathbb{R}^d$). For a given point x , let $R_k(x)$ denote the distance from x to its k -th nearest neighbor point.

The kNN density estimator estimates the density by

$$\hat{p}_{\text{knn}}(x) = \frac{k}{n} \cdot \frac{1}{V_d R_k^d(x)} = \frac{k}{n} \cdot \frac{1}{\text{Vol. of a } d\text{-dimensional ball with radius } R_k(x)}.$$

- $d = 1, V_1 = 2$:

$$\hat{p}_{\text{knn}}(x) = \frac{k}{n} \frac{1}{2R_k(x)}.$$

- $d = 2, V_2 = \pi$:

$$\hat{p}_{\text{knn}}(x) = \frac{k}{n} \frac{1}{\pi R_k^2(x)}.$$

kNN (Cont'd)

By the definition of $R_k(x)$, the ball centered at x with radius $R_k(x)$,

$$B(x, R_k(x)) = \{y : \|x - y\| \leq R_k(x)\},$$

satisfies

$$\frac{k}{n} = \frac{1}{n} \sum_{i=1}^n I(X_i \in B(x, R_k(x))).$$

Recall from the relation between EDF and CDF, the right-side of the equation can be viewed as an estimator of the quantity

$$P(X_i \in B(x, R_k(x))) \approx \int_{B(x, R_k(x))} p(y) dy,$$

where $p(y) \approx p(x)$ for every $y \in B(x, R_k(x))$. Thus,

$$P(X_i \in B(x, R_k(x))) \approx p(x) \int_{B(x, R_k(x))} dy = p(x) V_d R_k^d(x).$$

kNN (Cont'd)

We have

$$P(X_i \in B(x, R_k(x))) \approx p(x) V_d R_k^d(x).$$

This quantity will be the target of the estimator

$$\frac{1}{n} \sum_{i=1}^n I(X_i \in B(x, R_k(x))) = \frac{k}{n}.$$

This leads to

$$p(x) V_d R_k^d(x) \approx \frac{k}{n},$$

so

$$\hat{p}_{\text{knn}}(x) = \frac{k}{n} \frac{1}{V_d R_k^d(x)}.$$

Asymptotic theory in $d = 1$.

$$\text{bias}(\hat{p}_{\text{knn}}(x)) = b_1 \frac{p''(x)}{p^2(x)} \left(\frac{k}{n}\right)^2 + b_2 \frac{p(x)}{k} + o\left(\left(\frac{k}{n}\right)^2 + \frac{1}{k}\right),$$

where b_1 and b_2 are two constants.

$$\text{Var}(\hat{p}_{\text{knn}}(x)) = v_0 \cdot \frac{p^2(x)}{k} + o\left(\frac{1}{k}\right),$$

where v_0 is a constant. Therefore,

$$\text{MSE}(\hat{p}_{\text{knn}}(x)) = O\left(\frac{k^4}{n^4} + \frac{1}{k}\right)$$

which leads to $k = C_0 n^{4/5}$ for some constant C_0 .

Gaussian Mixture Model (GMM) and EM Algorithm

From Parametric Model to Mixture Model

Suppose we assume that the data follow a Gaussian distribution. Then the density estimator is

$$\hat{p}_n(x) = \frac{1}{\sqrt{2\pi\hat{\sigma}_n^2}} \exp \left\{ -\frac{(x - \hat{\mu}_n)^2}{2\hat{\sigma}_n^2} \right\}.$$

However, if the true density p is far from Gaussian,

$$\hat{p}_n(x) \xrightarrow{P} \bar{p}(x) \neq p(x).$$

Thus,

$$\hat{p}_n(x) - p(x) = \underbrace{\hat{p}_n(x) - \bar{p}(x)}_{\text{estimation error}} + \underbrace{\bar{p}(x) - p(x)}_{\text{model bias}}.$$

The model bias does not vanish even when $n \rightarrow \infty$.

Q. How can we make a parametric model more flexible?

A. Mixture of distributions

Gaussian Mixture Model

The Gaussian mixture model (GMM) uses

$$p_{\text{GMM}}(x) = \sum_{l=1}^K \pi_l \phi(x; \mu_l, \sigma_l^2)$$

where $\pi_l \geq 0$, $\sum_{l=1}^K \pi_l = 1$. Equivalently,

$$p_{\text{GMM}}(x) = \sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi\sigma_l^2}} \exp \left\{ -\frac{(x - \mu_l)^2}{2\sigma_l^2} \right\}.$$

Gaussian Mixture Model: Estimation

For each component, $(\pi_l, \mu_l, \sigma_l^2)$ is a set of three parameters, with one constraint $\sum_{l=1}^K \pi_l = 1$. Thus, there are totally $3K - 1$ parameters.

Estimation is often done by Maximum Likelihood Estimate(MLE):

$$\hat{\Theta} = \arg \max_{\Theta} \sum_{i=1}^n \log \left[\sum_{l=1}^K \pi_l \phi(X_i; \mu_l, \sigma_l^2) \right],$$

where $\Theta = \{(\pi_l, \mu_l, \sigma_l^2)\}_{l=1}^K$.

The final density estimator is

$$\hat{p}_{\text{GMM}}(x) = \sum_{l=1}^K \hat{\pi}_l \phi(x; \hat{\mu}_l, \hat{\sigma}_l^2).$$

Gaussian Mixture Model: Pros and Cons

Pros

- Each component can be interpreted as a hidden sub-population.
- π_l represents the proportion of the l -th sub-population.
- μ_l and σ_l^2 describe its center and variation.
- GMM can approximate complicated distributions when K is large.

Cons

- **Identifiability problem:** different parameter sets may produce the same distribution.
- **Computation problem:** the MLE usually has no closed-form solution.
- **Choice of K :** K acts as a tuning parameter and is difficult to choose.

Appendix: Expectation-Maximization (EM) Algorithm

Purpose:

The Expectation-Maximization (EM) algorithm is an iterative optimization technique for finding the MLE, especially when the model includes latent variables. For GMM, $\Theta = \{\pi_l, \mu_l, \sigma_l^2\}_{l=1}^K$.

Introduce a latent variable Z_i , where $Z_i = l$ means that observation X_i belongs to the l -th Gaussian component.

The observed-data log-likelihood is

$$\mathcal{L}(\Theta | X) = \sum_{i=1}^n \log \left(\sum_{l=1}^K \pi_l \phi(X_i; \mu_l, \sigma_l^2) \right).$$

The sum inside the logarithm makes direct maximization difficult.

Appendix: EM Algorithm: Core Idea

If the latent variables $Z = (Z_1, \dots, Z_n)$ were observed, the likelihood would be much simpler.

Instead of maximizing the observed-data log-likelihood directly, EM uses the expected complete-data log-likelihood

$$Q(\Theta, \Theta^{(t)}) = E_{Z|X, \Theta^{(t)}} [\log p(X, Z | \Theta)].$$

Starting with an initial parameter $\Theta^{(0)}$, EM repeatedly alternates between two steps:



until convergence.

Appendix: EM Algorithm: E-Step

The E-Step computes the posterior probability that observation X_i belongs to component l . This probability is called the *responsibility*:

$$\gamma_{il} = P(Z_i = l \mid X_i, \Theta^{(t)})$$

and

$$\gamma_{il} = \frac{\pi_l^{(t)} \phi \left(X_i; \mu_l^{(t)}, (\sigma_l^2)^{(t)} \right)}{\sum_{j=1}^K \pi_j^{(t)} \phi \left(X_i; \mu_j^{(t)}, (\sigma_j^2)^{(t)} \right)}.$$

Thus, γ_{il} represents the contribution of X_i to the estimation of the l -th component.

$$0 \leq \gamma_{il} \leq 1, \quad \sum_{l=1}^K \gamma_{il} = 1.$$

Appendix: EM Algorithm: M-Step

The M-Step maximizes $Q(\Theta, \Theta^{(t)})$ with respect to the new parameters:

$$\Theta^{(t+1)} = \arg \max_{\Theta} Q(\Theta, \Theta^{(t)}).$$

For GMM, this gives the update equations

$$\pi_l^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \gamma_{il}$$

$$\mu_l^{(t+1)} = \frac{\sum_{i=1}^n \gamma_{il} X_i}{\sum_{i=1}^n \gamma_{il}}$$

and

$$(\sigma_l^2)^{(t+1)} = \frac{\sum_{i=1}^n \gamma_{il} (X_i - \mu_l^{(t+1)})^2}{\sum_{i=1}^n \gamma_{il}}.$$

These new parameters are used in the next E-Step.

Appendix: Derivation of the EM Q-Function

For the complete data (X, Z) ,

$$p(X_i, Z_i \mid \Theta) = \prod_{l=1}^K [\pi_l \phi(X_i; \mu_l, \sigma_l^2)] I(Z_i = l).$$

Therefore,

$$\log p(X, Z \mid \Theta) = \sum_{i=1}^n \sum_{l=1}^K I(Z_i = l) [\log \pi_l + \log \phi(X_i; \mu_l, \sigma_l^2)].$$

The Q-function is

$$Q(\Theta, \Theta^{(t)}) = E_{Z|X, \Theta^{(t)}}[\log p(X, Z \mid \Theta)].$$

Appendix: Derivation of the EM Q-Function

Since

$$\mathbb{E}[I(Z_i = l) \mid X_i, \Theta^{(t)}] = P(Z_i = l \mid X_i, \Theta^{(t)}) = \gamma_{il},$$

we obtain

$$Q(\Theta, \Theta^{(t)}) = \sum_{i=1}^n \sum_{l=1}^K \gamma_{il} [\log \pi_l + \log \phi(X_i; \mu_l, \sigma_l^2)].$$

Thus, the difficult structure

$$\log \left(\sum_{l=1}^K \dots \right)$$

in the observed likelihood is replaced by

$$\sum_{l=1}^K \gamma_{il} \log(\dots),$$

which is much easier to maximize in the M-Step.

Appendix: EM Algorithm: Interpretation

The EM algorithm can be interpreted as repeatedly performing

soft assignment $\longrightarrow$ parameter update
--

- **E-Step:** Estimate how likely each observation belongs to each component.
- **M-Step:** Re-estimate the Gaussian parameters using these probabilities as weights.

For example,

$$\mu_l^{(t+1)} = \frac{\sum_i \gamma_{il} X_i}{\sum_i \gamma_{il}}$$

is simply a weighted sample mean.

The procedure is repeated until the parameters change very little.

Appendix: EM Algorithm: Properties

- **Monotonicity:**

The observed log-likelihood does not decrease:

$$\mathcal{L}(\Theta^{(t+1)} | X) \geq \mathcal{L}(\Theta^{(t)} | X).$$

- **Local convergence:**

EM may converge to a local maximum or a saddle point.

- **Initialization dependence:**

The final result can strongly depend on the initial parameter $\Theta^{(0)}$.

- Multiple random initializations may therefore be useful.
- EM can converge slowly in some problems.

Regression Trees and Classification

Regression Tree

Assume the covariate may have multiple dimensions, i.e.,

$$x = (x_1, \dots, x_d).$$

And our data

$$(X_1, Y_1), \dots, (X_n, Y_n) \sim P$$

for some CDF P .

Regression tree constructs an estimator of the form

$$m(x) = \sum_{l=1}^M c_l I(x \in R_l)$$

where R_l is some rectangular partition of the space of covariates.

Regression Tree (Example)

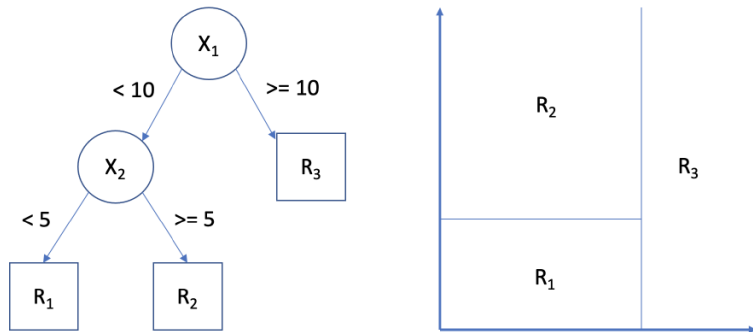


Figure: Two covariates ($d = 2$) and three regions R_1, R_2, R_3 .

Regression Tree

A regression tree estimator will predict the same value of the response Y within the same area of the covariate.

$\implies m(x)$ will be the same when x is within the same area.

There are $2M$ quantities to be determined:

- 1 the regions $R_1, \dots, R_M$,
- 2 the predicted values $c_1, \dots, c_M$.

Given regions,

$$\hat{c}_l = \frac{\sum_{i=1}^n Y_i I(X_i \in R_l)}{\sum_{i=1}^n I(X_i \in R_l)}$$

A challenging part is how to determine

$$R_1, \dots, R_M.$$

Regression Tree (Algorithm)

Let X_{ij} be the j -th coordinate of the i -th observation (X_i).

- 1 For a given j , define

$$R_a(j, s) = \{x : x < s\}, \quad R_b(j, s) = \{x : x \geq s\}.$$

- 2 Find c_a and c_b that minimize

$$\sum_{X_i \in R_a} (Y_i - c_a)^2, \quad \sum_{X_i \in R_b} (Y_i - c_b)^2,$$

respectively.

- 3 Compute the score

$$S(j, s) = \sum_{X_i \in R_a} (Y_i - c_a)^2 + \sum_{X_i \in R_b} (Y_i - c_b)^2.$$

Regression Tree (Algorithm)

- 4 Change s and repeat the same calculation until we find the minimizer of $S(j, s)$ denoted by $S^*(j)$.
- 5 Compute the score $S^*(j)$ for $j = 1, \dots, d$.
- 6 Pick the dimension (coordinate) and the corresponding split point s that has the minimal score $S^*(j)$. Partition the space into two parts according to this split.
- 7 Repeat the above procedure for each partition until certain stopping criterion is satisfied.

With a collection of rectangle partitions $\hat{R}_1, \dots, \hat{R}_M$, the final estimator is

$$\hat{m}(x) = \sum_{l=1}^M \hat{c}_l I(x \in \hat{R}_l).$$

A tree can be grown by repeatedly reducing the SSE. To avoid overfitting, we later prune the tree using a criterion that balances fitting error and tree complexity.

Basic terms for Classification

Here, we consider a simple scenario: **binary classification**. Given a feature vector x_0 , we want to create a classifier c that maps

$$c : x_0 \mapsto \{0, 1\}.$$

We measure the classification error using a loss function L . For example, the 0–1 loss is

$$L(c(x_0), y_0) = I(c(x_0) \neq y_0).$$

The risk function is the expected loss:

$$R(c) = E[L(c(X), Y)].$$

Our goal is to find a classifier that minimizes the risk.

Bayes Classifier

Under the 0–1 loss, an optimal classifier is

$$c^*(x) = \arg \max_{y=0,1} P(Y = y \mid X = x).$$

Equivalently,

$$c^*(x) = \begin{cases} 0, & P(Y = 0 \mid X = x) \geq P(Y = 1 \mid X = x), \\ 1, & P(Y = 1 \mid X = x) > P(Y = 0 \mid X = x). \end{cases}$$

This classifier minimizes the risk: $R(c^*) = \min_c R(c)$. It is called the **Bayes classifier**. Since $Y \in \{0, 1\}$,

$$\mathbb{E}(Y \mid X = x) = P(Y = 1 \mid X = x).$$

Thus, a regression estimator can be converted into a classifier.

Nonparametric Classification

- We can construct an optimal classifier if we know the true underlying probability model.
- Even if we do not know the true model, we can estimate the model nonparametrically and then train a classifier using the estimated density function, or regression function.
- However, this approach can be practically problematic because the dimension of the data is often large.
- Note that converting a nonparametric estimator to a classifier may not work well in practice.

Instead, we can construct a classifier more directly.

k-NN Approach

We are interested in the class label at the two blue boxes x_1 and x_2 .

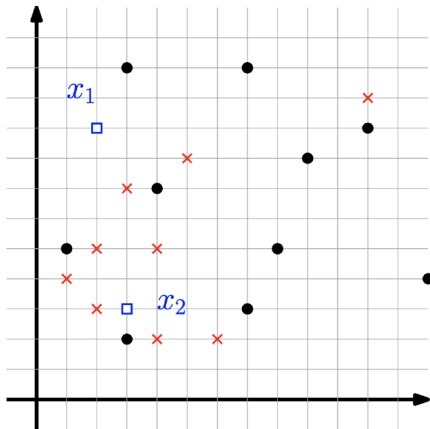


Figure: Two classes: black dots and red crosses.

k-NN Approach

The idea is very simple.

- For a given point x_0 , we find its k nearest data points.
- We compare the labels of these k points.
- We assign the label of x_0 as the majority label among these k points.

For example, assume we use a 3-NN classifier $\hat{c}_{3\text{-NN}}$.

$$\hat{c}_{3\text{-NN}}(x_1) = \text{black dot},$$

$$\hat{c}_{3\text{-NN}}(x_2) = \text{red cross}.$$

If there are ties, we randomly assign one of the class labels attaining the tie.

- The k-NN approach is simple and easy to operate.
- It can be easily generalized to multiple classes.
- The choice of k is often done by cross-validation.

Decision Tree

Decision tree is another common approach in classification. A decision tree is like a regression tree:

Partition the covariate space into rectangular regions

and assign each region a class label. The class label at each region is determined by **majority vote**.

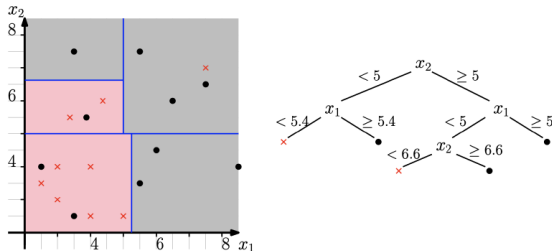


Figure: (left) the scatterplot and regions separated by a decision tree; (right) the tree structure.

Decision Tree

In the case of binary classification,

- Let $(X_1, Y_1), \dots, (X_n, Y_n)$ denote the data.
- Let $R_1, \dots, R_k$ denote a rectangular partition of the space of the covariates.
- Let $R(x)$ denote the rectangular region where x falls within.

The decision tree is

$$\hat{c}_{DT}(x) = I \left(\frac{\sum_{i=1}^n Y_i I(X_i \in R(x))}{\sum_{i=1}^n I(X_i \in R(x))} > \frac{1}{2} \right).$$

Thus, the label is 1 whenever the average response in the region containing x is greater than $1/2$.

The decision tree is essentially a classifier converted from a regression tree!

Appendix: Proofs

Proof: Bias of the Histogram Estimator

For $x \in B_l$, recall that

$$\hat{p}_n(x) = \frac{M}{n} \sum_{i=1}^n I(X_i \in B_l).$$

Therefore,

$$\mathbb{E}[\hat{p}_n(x)] = MP(X_i \in B_l) = M \int_{(l-1)/M}^{l/M} p(t) dt.$$

By the Mean Value Theorem for integrals, there exists $x^* \in \left[\frac{l-1}{M}, \frac{l}{M} \right]$ such that $\mathbb{E}[\hat{p}_n(x)] = p(x^*)$. Hence, $\text{Bias}(\hat{p}_n(x)) = p(x^*) - p(x)$. Since $|p'(x)| \leq L$,

$$|p(x^*) - p(x)| \leq L|x^* - x| \leq \frac{L}{M}.$$

Thus, $|\text{Bias}(\hat{p}_n(x))| \leq \frac{L}{M}$.

Proof: Variance of the Histogram Estimator

For $x \in B_I$, let $\pi_I = P(X_i \in B_I)$. Then

$$\sum_{i=1}^n I(X_i \in B_I) \sim \text{Binomial}(n, \pi_I).$$

Therefore,

$$\text{Var}(\hat{p}_n(x)) = \frac{M^2}{n^2} \text{Var} \left(\sum_{i=1}^n I(X_i \in B_I) \right) = \frac{M^2}{n^2} n \pi_I (1 - \pi_I).$$

From the previous slide,

$$\pi_I = \int_{(I-1)/M}^{I/M} p(t) dt = \frac{p(x^*)}{M}.$$

Thus,

$$\text{Var}(\hat{p}_n(x)) = \frac{M^2}{n} \frac{p(x^*)}{M} \left(1 - \frac{p(x^*)}{M} \right) = M \frac{p(x^*)}{n} - \frac{p^2(x^*)}{n}.$$

Proof: Bias of the KDE

Recall that

$$\hat{p}_n(x_0) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right).$$

Then,

$$\begin{aligned}\mathbb{E}[\hat{p}_n(x_0)] - p(x_0) &= \frac{1}{h} E\left[K\left(\frac{X_i - x_0}{h}\right)\right] - p(x_0) \\ &= \frac{1}{h} \int K\left(\frac{x - x_0}{h}\right) p(x) dx - p(x_0).\end{aligned}$$

Let $y = \frac{x - x_0}{h}$. Then $x = x_0 + hy$, $dx = h dy$, so

$$\mathbb{E}[\hat{p}_n(x_0)] - p(x_0) = \int K(y) p(x_0 + hy) dy - p(x_0).$$

Proof: Bias of the KDE (Cont'd)

By Taylor's expansion,

$$p(x_0 + hy) = p(x_0) + hy p'(x_0) + \frac{1}{2}h^2 y^2 p''(x_0) + o(h^2).$$

Therefore,

$$\begin{aligned}\mathbb{E}[\hat{p}_n(x_0)] - p(x_0) &= p(x_0) \int K(y) dy + hp'(x_0) \int yK(y) dy \\ &\quad + \frac{1}{2}h^2 p''(x_0) \int y^2 K(y) dy + o(h^2) - p(x_0).\end{aligned}$$

Since $\int K(y) dy = 1$ and, by symmetry of K , $\int yK(y) dy = 0$, define $\mu_K = \int y^2 K(y) dy$. Hence,

$$\mathbb{E}[\hat{p}_n(x_0)] - p(x_0) = \frac{1}{2}h^2 p''(x_0)\mu_K + o(h^2).$$

Thus, $\text{Bias}(\hat{p}_n(x_0)) = O(h^2)$.

Proof: Variance of the KDE

By independence,

$$\text{Var}(\hat{p}_n(x_0)) = \frac{1}{nh^2} \text{Var} \left[K \left(\frac{X_i - x_0}{h} \right) \right].$$

Then,

$$\text{Var}(\hat{p}_n(x_0)) \leq \frac{1}{nh^2} E \left[K^2 \left(\frac{X_i - x_0}{h} \right) \right].$$

Thus,

$$\text{Var}(\hat{p}_n(x_0)) \leq \frac{1}{nh^2} \int K^2 \left(\frac{x - x_0}{h} \right) p(x) dx.$$

Again, let $y = \frac{x - x_0}{h}$. Then

$$\text{Var}(\hat{p}_n(x_0)) \leq \frac{1}{nh} \int K^2(y) p(x_0 + hy) dy.$$

Proof: Variance of the KDE (Cont'd)

By Taylor's expansion,

$$p(x_0 + hy) = p(x_0) + hy p'(x_0) + o(h).$$

Thus,

$$\text{Var}(\hat{p}_n(x_0)) \leq \frac{1}{nh} \int K^2(y) [p(x_0) + hy p'(x_0) + o(h)] dy.$$

Since K^2 is symmetric, $\int yK^2(y) dy = 0$. Therefore,

$$\text{Var}(\hat{p}_n(x_0)) \leq \frac{p(x_0)}{nh} \int K^2(y) dy + o\left(\frac{1}{nh}\right).$$

Let $\sigma_K^2 = \int K^2(y) dy$. Hence,

$$\text{Var}(\hat{p}_n(x_0)) \leq \frac{p(x_0)\sigma_K^2}{nh} + o\left(\frac{1}{nh}\right).$$

In particular, $\text{Var}(\hat{p}_n(x_0)) = O\left(\frac{1}{nh}\right)$.

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Thank you!